Dupire PINN / PDF validation

The risk-neutral density and the $K=0$ boundary: derivation, observation, diagnosis

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Outline

The risk-neutral density

December observations: a deviation

The NN density in scaled coordinates

IBP sanity checks

What went wrong

Setup: from option prices to a density

$$C(K,T) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] = e^{-rT}\int_K^{\infty} (S-K)\,f(S)\,dS$$

- f is the (unknown) risk-neutral density of S_T. Goal: recover f from prices, then check it against Monte-Carlo simulation of the calibrated model.
- Our NN parameterizes C; the implied density follows by differentiation (next slide).

Breeden–Litzenberger: the density is $\partial^2 C/\partial K^2$

Differentiate under the integral (Leibniz; the lower-limit boundary term vanishes since the integrand $(S - K)$ is zero at $S = K$):

$$\frac{\partial C}{\partial K} = -e^{-rT} \int_K^\infty f(S)\, dS = -e^{-rT}\big(1 - F(K)\big) = -e^{-rT}\, \mathbb{Q}(S_T > K),$$

$$\frac{\partial^2 C}{\partial K^2} = e^{-rT}\, f(K).$$

$$\boxed{q(K, T) \;:=\; e^{rT}\, \frac{\partial^2 C}{\partial K^2}(K, T) \;=\; f_{\mathbb{Q}}(K)}$$

Regularity: f integrable, $C \to 0$ and $\partial_K C \to 0$ as $K \to \infty$.

The mean identity (integration by parts)

$$\mathbb{E}[S_T] = \int_0^\infty K\, q(K,T)\, dK = e^{rT} \int_0^\infty K\, \frac{\partial^2 C}{\partial K^2}\, dK$$

IBP:

$$\int_0^\infty K\, \frac{\partial^2 C}{\partial K^2}\, dK = \left[K\, \frac{\partial C}{\partial K} - C \right]_0^\infty = C(0,T) = S_0$$

(at ∞: $C \to 0$, $K\partial_K C \to 0$; at 0: $K\partial_K C \to 0$ and $C(0,T) = S_0$, the zero-strike call equals spot).

$$\boxed{\mathbb{E}[S_T] = e^{rT}\, C(0,T) = S_0\, e^{rT}}$$

Three independent estimators that must all equal the forward $M_{\mathrm{ref}} = S_0 e^{rT}$:

- $M_1 = e^{rT} \int_0^\infty K\, \partial_K^2 C\, dK$ (density quadrature),
- $M_2 = e^{rT}\, C(0,T)$ (closed form),
- $M_3 = \mathbb{E}[S_T]$ (Monte Carlo).

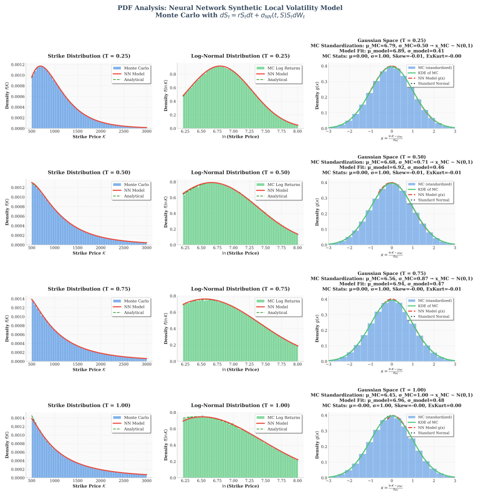
What we plot

For each maturity T, three panels:

1. Strike-space density $f(K)$ — MC histogram vs NN model vs analytical.
2. Log-strike density $f(\ln K)$.
3. Gaussian space — standardized MC (KDE) vs NN, against $\mathcal{N}(0,1)$.

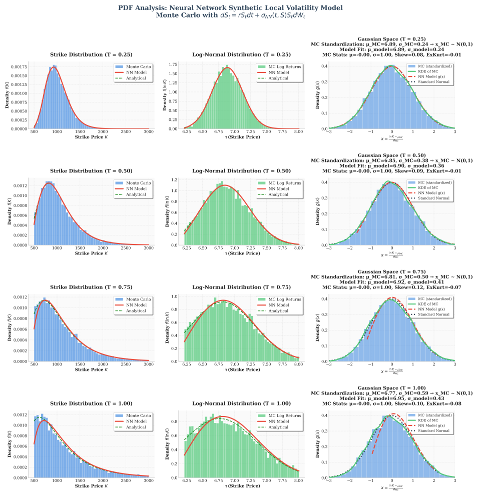
The NN curve is the Breeden–Litzenberger density $e^{rT}\partial_K^2 C_{\mathrm{NN}}$; the MC histogram is the simulated S_T under the calibrated σ_{NN}.

Control: constant volatility $\sigma = 1$



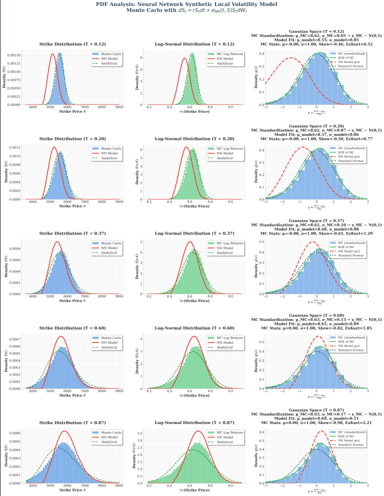
$T \in \{0.25, 0.5, 0.75, 1.0\}$. NN, MC and the log-normal analytical density are essentially indistinguishable — the clean baseline, no deviation.

Synthetic Dupire volatility



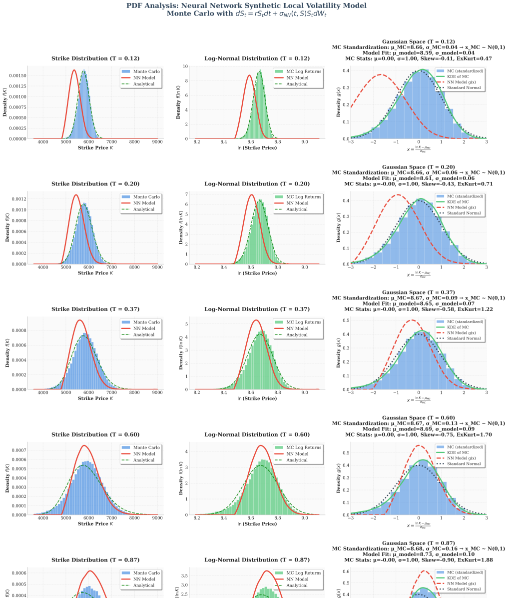
$\sigma(x,t) = 0.3 + ye^{-y}$, $y = (t+0.1)\sqrt{x+0.1}$. NN tracks MC closely; a small deficit is already visible in the Gaussian panel.

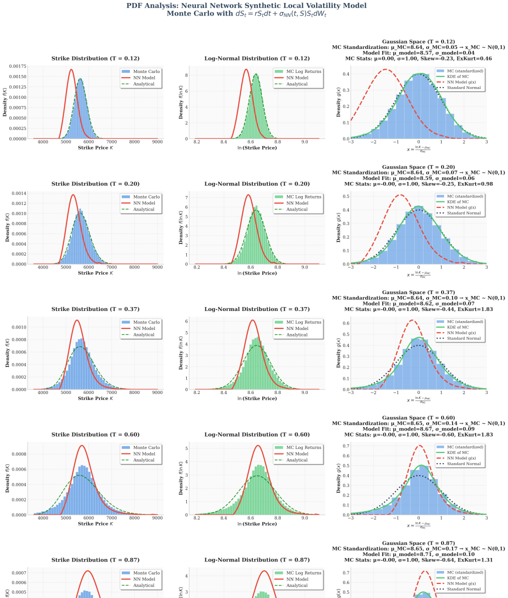
Market data: DAX 7 Aug 2001



$K \in [4000, 9000]$, $T \in \{0.12, 0.20, 0.37, 0.60, 0.87\}$. Clear NN-vs-MC separation in strike space; in Gaussian space MC tracks $\mathcal{N}(0,1)$ but the NN density deviates — the $K=0$ boundary fingerprint.

DAX 8–9 Aug 2001: consistent across days





The scaling the network uses

$$t = \frac{T}{T_{\max}}, \qquad k = \frac{e^{-rT}K}{K_{\max}}, \qquad \tilde{\varphi} = \frac{C}{S_0}, \qquad \eta = \tfrac{1}{2}\, T_{\max}\, \sigma^2$$

Price ansatz (as trained):

$$\tilde{\varphi}(k,t) = 1 - e^{-\mathcal{N}_c(k,t)}, \qquad \mathcal{N}_c = \mathrm{NN}_{\varphi} \text{ (softplus output, } > 0).$$

Rederiving the density under the scaling

The map $K \mapsto k$ is linear, so the Jacobian is constant:

$$\frac{dk}{dK} = \frac{e^{-rT}}{K_{\max}}, \qquad \frac{d^2k}{dK^2} = 0.$$

Hence (no curvature term):

$$\frac{\partial^2 C}{\partial K^2} = S_0 \left(\frac{dk}{dK}\right)^2 \frac{\partial^2 \tilde{\varphi}}{\partial k^2}.$$

$$\boxed{q(K,T) = e^{rT}\frac{\partial^2 C}{\partial K^2} = e^{rT} S_0 \left(\frac{e^{-rT}}{K_{\max}}\right)^2 \frac{\partial^2 \tilde{\varphi}}{\partial k^2} = \frac{S_0\, e^{-rT}}{K_{\max}^2} \frac{\partial^2 \tilde{\varphi}}{\partial k^2}}$$

Scaled Dupire PDE: $\partial_t \varphi = \eta(k,t)\, k^2\, \partial_k^2 \varphi$.

Implementation verified

- This boxed formula is exactly what `_raw_model_density` computes (double autodiff for $\partial_k^2 \tilde{\varphi}$, Jacobian $(e^{-rT}/K_{\max})^2$, $\times S_0$, $\times e^{rT}$).
- The density and IBP-diagnostic code (`_raw_model_density`, `compute_mean_diagnostics`, `prepare_data_for_nn`, `phi_tilde_from_nn`) were audited line-by-line against the math above — Jacobian power, a single e^{rT}, the S_0 factor — and all match.
- Numerically re-checked: the autodiff second derivative and the trapezoidal quadrature are mutually consistent — $M_1 \equiv M_2$ to machine precision (the IBP identity; see next section).

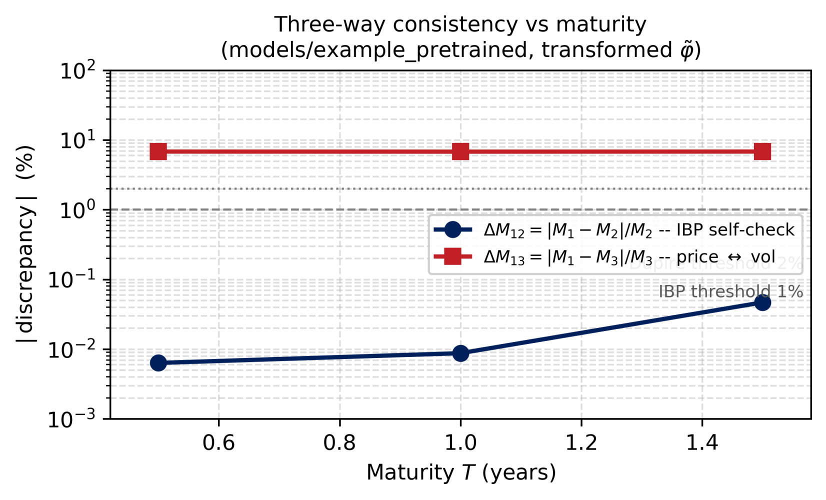
The code is correct; the residual the diagnostic exposes is a property of the trained *model*, not of its implementation.

Three-way check (pretrained, synthetic Dupire surface)

T	M_{ref}	M_1	%err	M_2	%err	M_3	%err	$\Delta M_{12}\%$	$\Delta M_{13}\%$
0.5	1020.20	948.87	−6.99	948.93	−6.99	1018.25	−0.19	0.006	6.814
1.0	1040.81	967.66	−7.03	967.74	−7.02	1038.19	−0.25	0.009	6.794
1.5	1061.84	982.49	−7.47	982.95	−7.43	1054.36	−0.70	0.047	6.816

- $M_1 \equiv M_2$ to $< 0.05\%$: autodiff + quadrature consistent (IBP identity exact).
- M_3 within 0.7% of forward: NN_η (the vol net) is well-calibrated.
- $M_1, M_2 \approx 7\%$ below forward: NN_φ (the price net) undershoots near $K = 0$.

ΔM_{12}, ΔM_{13} vs maturity

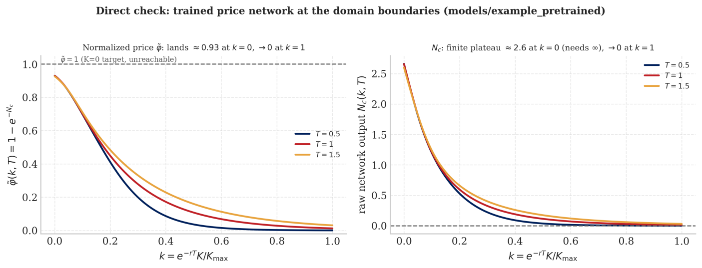


ΔM_{12} at machine precision (IBP exact); $\Delta M_{13} \approx 7\%$, flat in T — a systematic price-net gap, not a calibration drift.

Direct check: NN output at the boundaries

Evaluating the trained network at $k = 0$ and $k = 1$:

T	$\mathcal{N}_c(0, T)$	$\tilde{\varphi}(0, T)$	deficit	$\mathcal{N}_c(1, T)$	$\tilde{\varphi}(1, T)$
0.5	2.661	0.9301	6.99%	0.0007	0.0007
1.0	2.656	0.9298	7.02%	0.0128	0.0128
1.5	2.600	0.9257	7.43%	0.0315	0.0310



$\mathcal{N}_c(0, T) \approx 2.6$ (finite — would need ∞); $\mathcal{N}_c(1, T) \approx 0.01$. $e^{rT} C_{\mathrm{NN}}(0, T)$ reproduces $M_2 = 948.9/967.7/983.0$.

Counterfactual: if the $(1-k)$ factor were present

Applying eq. 3.9's $(1-k)$ factor to the trained $\mathcal{N}_c$ at analysis time:

T	M_2	M_1^{cap}	M_1 (pred)	$\Delta M_{12}^{\mathrm{cap}}\%$
0.5	948.93	948.25	948.22	0.072
1.0	967.74	954.37	954.42	1.381
1.5	982.95	949.51	949.50	3.402

- $k=0$: M_2 **unchanged** (factor $=1$ there) — the 7% deficit is identical.
- $k=1$: $\tilde{\varphi}$ forced to **exactly 0** by construction.
- But $M_1 \neq M_2$: a real boundary-slope gap $= e^{rT} S_0 \mathcal{N}_c(1)$, growing 0.07% $\to$ 1.4% $\to$ 3.4% with T. Verified numerically (matches $M_2 - e^{rT} S_0 \mathcal{N}_c(1)$).

The output map cannot reach $K = 0$

$$\tilde{\varphi} = 1 - e^{-\mathcal{N}_c} \in (0, 1), \qquad \mathcal{N}_c = \text{softplus} > 0 \;\Rightarrow\; \text{range open at } 1.$$

- $K = \infty$ ($k = 1$): BC $\tilde{\varphi} = 0$ needs $\mathcal{N}_c = 0$ — **attainable** (measured ≈ 0.01).
- $K = 0$ ($k = 0$): BC $\tilde{\varphi} = 1$ needs $\mathcal{N}_c \to \infty$ — **unattainable** by a finite output (measured ≈ 2.6). So $C_{\mathrm{NN}}(0, T) < S_0$ by construction; the gap is the 7%.

Why the deficit is frozen in T

Scaled Dupire PDE:

$$\frac{\partial \varphi}{\partial t} = \eta(k,t)\, k^2 \frac{\partial^2 \varphi}{\partial k^2}.$$

At $k = 0$ the k^2 factor annihilates the RHS, so $\partial_t \varphi(0,t) = 0 \Rightarrow \varphi(0,t) = \varphi(0,0)\ \forall t$.

The $T = 0$ initial-condition deficit (which a finite ansatz cannot avoid) is propagated **unchanged** to every maturity — exactly the flat $\Delta M_{13} \approx 7\%$.

Summary: the $(1-k)$ factor is not the fix

- The deviation seen in December is the NN_φ $K{=}0$ undershoot, localized by the IBP three-way test; NN_η is fine (M_3 on the forward).
- The manuscript's $(1-k)$ factor only pins $K=\infty$; at $k=0$ it is 1, so the 7% deficit is identical — and it introduces a new $M_1 \neq M_2$ gap.
- Fixes: a $K{=}0$ penalty $\lambda_{k_0}|C(0,T)-S_0|^2$ in the loss, or an ansatz that can attain $\tilde{\varphi}=1$, e.g. the parity form $\pi^c = S_0 - Ke^{rT}\left[1-(1-k)e^{-\mathcal{N}_c}\right]$ (pins *both* ends by construction).

Questions?

Plots: `plots/ibp_comparison/`